

Kinematic Classification of Faults: Oblique-Slip Faults, Net-Slip Vectors & Pitch/Rake Analysis

Comprehensive Geological Concept Guide · Authoritative References: Fossen (2010, Ch. 9) & Van der Pluijm & Marshak (2004, Ch. 8)

THEORETICAL FOUNDATION: KINEMATIC GEOMETRY & FAULT SLIP DYNAMICS

Displacement Vectors, Components of Slip, Pitch (Rake) Measurement & The Oblique-Slip Spectrum

1. The Kinematic Definition of Fault Slip

A **fault** is a planar or subplanar discontinuity across which measurable displacement has occurred by brittle or semi-brittle shear fracture. To rigorously characterize the kinematics of fault movement, structural geologists track the relative displacement between points that were originally contiguous across the fault surface before deformation:

Net Slip ($\vec{S}$): The total vector displacement connecting two points on opposite sides of the fault that were once directly adjacent (piercing points). The net slip vector lies entirely within the fault plane.

Because the fault plane is generally inclined at a dip angle (δ) and has a specific strike direction, the net slip vector $\vec{S}$ can be oriented in any arbitrary direction on that plane. The orientation of the slip vector within the fault plane is quantified by its **pitch (or rake)**.

VECTOR DECOMPOSITION OF NET SLIP

$$\vec{S}_{\text{net}} = \vec{S}_{\text{strike-slip}} + \vec{S}_{\text{dip-slip}}$$

$$|S_{\text{ss}}| = S \cdot \cos(\phi) \quad (\text{Strike-slip component parallel to fault strike})$$

$$|S_{\text{ds}}| = S \cdot \sin(\phi) \quad (\text{Dip-slip component parallel to true dip direction})$$

$$S = \sqrt{S_{\text{ss}}^2 + S_{\text{ds}}^2}$$

Where:

- ϕ = Pitch (rake) angle of striations/slickenlines on the fault plane ($0^\circ \leq \phi \leq 90^\circ$)
- δ = True dip of the fault plane ($0^\circ < \delta \leq 90^\circ$)

Further projecting the dip-slip component into spatial coordinates yields the standard engineering and mining parameters:

- **Throw:** The vertical component of dip slip: $\text{Throw} = S_{\text{ds}} \cdot \sin(\delta) = S \cdot \sin(\phi) \cdot \sin(\delta)$.
- **Heave:** The horizontal component of dip slip perpendicular to the fault strike: $\text{Heave} = S_{\text{ds}} \cdot \cos(\delta) = S \cdot \sin(\phi) \cdot \cos(\delta)$.

*"Many faults show some deviation from true dip-slip and strike-slip displacement in the sense that the net slip vector is oblique. Such faults are called **oblique-slip faults**. The degree of obliquity is given by the pitch (also called rake), which is the angle between the strike of the slip surface and the slip vector (striation)."*

— Haakon Fossen, *Structural Geology* (2010), Chapter 9: Faults, §9.1 Fault Terminology, p. 153.

2. Textbook Visual Evidence: Oblique-Slip Geometry in Block Diagrams

Figure 9.6 from Fossen demonstrates how an oblique-slip fault displaces tilted stratigraphic layers. Because the net displacement vector is neither horizontal nor parallel to the dip, it simultaneously offsets markers both laterally across strike and vertically:

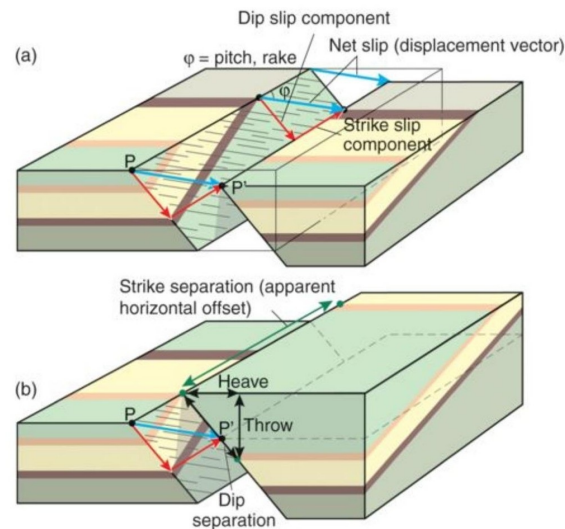


Figure 9.6 Illustration of an oblique-slip fault affecting tilted layers. The fault is normal sinistral and the blue arrows show the true displacement or net slip, which has a dip-slip and a strike-slip component (a). The net slip vector connects points (for example, P and P' in the figure) that were adjacent to each other before faulting. In (b) we have added stratigraphy to the hanging wall up to the level of the footwall. Now the strike and dip separation of the layers, which depend on the orientation of the layers and the nature of the fault displacement,

Figure 1 (Fossen Fig. 9.6): Block diagram of an oblique-slip fault displacing tilted sedimentary layers. The net displacement vector ($\vec{D}$ or net slip $\vec{S}$) is inclined obliquely on the fault plane. The angle ϕ (ϕ) is the *pitch* (*rake*) between the horizontal strike line and the slip vector. In this specific illustration, the hanging wall has moved downward (normal component) and to the left (sinistral strike-slip component), classifying it as a **sinistral-normal oblique-slip fault**. (Source: Haakon Fossen, *Structural Geology*, 2010, Cambridge University Press, Chapter 9).

*"If the net-slip vector parallels the dip direction of the fault (within $\sim 10^\circ$), the fault is called a **dip-slip fault**; if the vector roughly parallels the strike of the fault, the fault is called a **strike-slip fault**. If the vector is not parallel to either dip direction or the strike, we call the fault an **oblique-slip fault**. Oblique-slip faults have both a strike-slip and a dip-slip component of movement."*

— B. A. van der Pluijm & S. Marshak, *Earth Structure: An Introduction to Structural Geology and Tectonics* (2nd Ed., 2004), Chapter 8, p. 170.

3. The Classification Spectrum: Dip vs. Pitch (Angelier Diagram)

In nature, faults rarely plot as pure kinematic end-members. Instead, faults occupy a continuous spectrum based on the fault plane dip (δ) and the rake/pitch (ϕ) of the slickenlines:

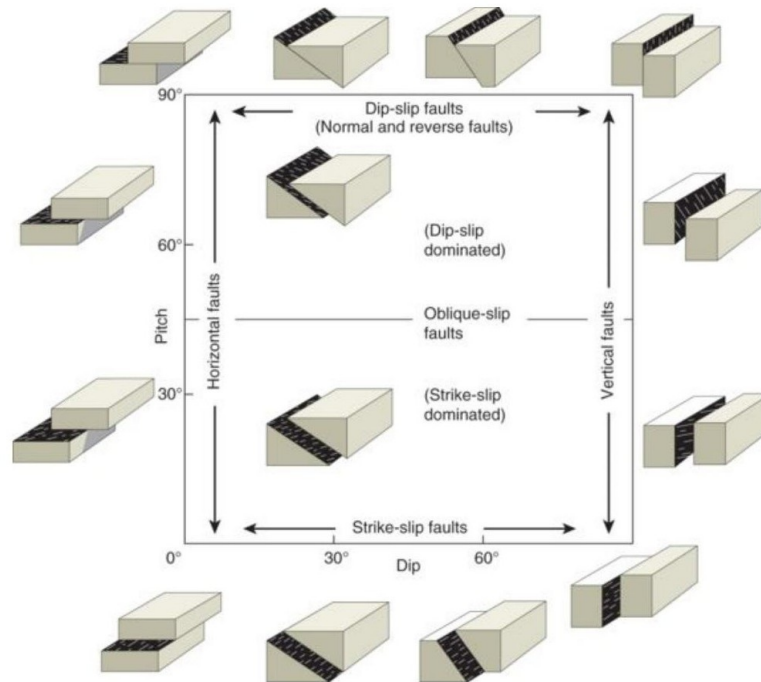


Figure 9.8 Classification of faults based on the dip of the fault plane and the pitch, which is the angle between the slip direction (displacement vector) and the strike.

Based on Angelier (1994).

Figure 2 (Fossen Fig. 9.8): Kinematic classification diagram of faults based on the dip of the fault plane (0° – 90° , x-axis) and the pitch/rake angle of the slip vector (0° – 90° , y-axis), adapted from Angelier (1994). Pure dip-slip faults (normal and reverse/thrust) plot along the top margin ($\phi \approx 90^\circ$). Pure strike-slip faults plot along the baseline ($\phi \approx 0^\circ$). The entire expansive intermediate domain (typically $10^\circ \leq \phi \leq 80^\circ$) represents **oblique-slip faults**.

Kinematic Classification Boundaries

Kinematic Category	Pitch / Rake (ϕ)	S_{ss} vs. S_{ds} Ratio	Dominant Motion & Slickenline Orientation
Pure Strike-Slip Fault	$0^\circ \leq \phi < 10^\circ$	$ S_{ss} \gg S_{ds} $ ($S_{ds} \approx 0$)	Horizontal slip parallel to strike; horizontal slickenlines/grooves.
Oblique-Slip (Strike-slip dominant)	$10^\circ \leq \phi < 45^\circ$	$ S_{ss} > S_{ds} > 0$	Both components significant; lateral shear exceeds vertical offset.
Oblique-Slip (Dip-slip dominant)	$45^\circ \leq \phi \leq 80^\circ$	$ S_{ds} > S_{ss} > 0$	Both components significant; vertical throw/heave exceeds lateral offset.
Pure Dip-Slip Fault	$80^\circ \leq \phi \leq 90^\circ$	$ S_{ds} \gg S_{ss} $ ($S_{ss} \approx 0$)	Down-dip or up-dip movement; slickenlines plunge down true dip.

4. The Four Subtypes of Oblique-Slip Faults

Because oblique-slip faults combine one of two vertical motions (normal vs. reverse) with one of two lateral motions (dextral vs. sinistral), there are exactly four fundamental geometric combinations:

Subtype Name	Dip-Slip Component	Strike-Slip Component	Typical Geodynamic Setting
Dextral-Normal (Right-lateral Normal)	Hanging wall moves downward relative to footwall	Opposite block moves rightward to observer	Transtensional rifts, oblique continental rifting (e.g., Gulf of California, Dead Sea pull-apart basins).
Sinistral-Normal (Left-lateral Normal)	Hanging wall moves downward relative to footwall	Opposite block moves leftward to observer	Transtensional shear zones, rift-flank stepovers, Basin and Range accommodation zones.
Dextral-Reverse (Right-lateral Reverse)	Hanging wall moves upward relative to footwall	Opposite block moves rightward to observer	Transpressional orogens, oblique subduction margins (e.g., Alpine Fault NZ, San Andreas Transverse Ranges).
Sinistral-Reverse (Left-lateral Reverse)	Hanging wall moves upward relative to footwall	Opposite block moves leftward to observer	Transpressional continental collision belts (e.g., Altyn Tagh flank thrusts, Zagros crush zone).

Transtension vs. Transpression:

- **Transtension:** Non-coaxial deformation combining horizontal strike-slip shear with crustal extension (yielding normal oblique-slip faulting, negative flower structures, and rhomb-shaped pull-apart basins).
- **Transpression:** Non-coaxial deformation combining horizontal strike-slip shear with crustal shortening (yielding reverse oblique-slip faulting, positive flower structures / palm-tree structures, and en-echelon push-up ranges).



EXAM QUESTION & RIGOROUS OPTION-BY-OPTION DISSECTION

DGMS 2019 Examination · Official Question Statement, Verified Answer & Detailed Distracter Analysis

DGMS 2019 · QUESTION 15

A fault having both dip-slip as well as strike-slip components is termed as:

- (a) Strike-slip fault
- (b) Dip-slip fault
- (c) Oblique-slip fault
- (d) Wrench fault



Correct Option: (c) Oblique-slip fault

Verified via Van der Pluijm & Marshak (Earth Structure, p. 170) & Haakon Fossen (Structural Geology, p. 153)

INCORRECT Option (a): Strike-slip fault

Detailed Refutation: By standard structural definition, a *strike-slip fault* (or transcurrent fault) is an end-member structure in which displacement is parallel to the strike of the fault surface. The net slip vector is entirely horizontal ($S_{\text{net}} \approx S_{\text{ss}}$), and the dip-slip component is zero ($S_{\text{ds}} = 0$, pitch $\phi \approx 0^\circ$). If a fault possesses a substantial dip-slip component alongside strike-slip, it cannot be classified as a pure strike-slip fault.

INCORRECT Option (b): Dip-slip fault

Detailed Refutation: A *dip-slip fault* is an end-member structure where the net slip vector is oriented parallel to the true dip line of the fault plane. Motion is strictly up-dip (reverse/thrust) or down-dip (normal), with slickenlines plunging at $\phi \approx 90^\circ$. By definition, a pure dip-slip fault has zero strike-slip component ($S_{\text{ss}} = 0$). Hence, this option contradicts the question's premise of possessing both components.

CORRECT ANSWER **Option (c): Oblique-slip fault**

Detailed Validation: An **oblique-slip fault** is explicitly defined in structural geology as a fault on which the slip direction is neither purely horizontal (strike-parallel) nor purely vertical/down-dip (dip-parallel), but intermediate between the two ($0^\circ < \phi < 90^\circ$, typically $10^\circ \leq \phi \leq 80^\circ$). Consequently, its net slip vector decomposes into non-zero orthogonal strike-slip and dip-slip components ($S_{\text{ss}} \neq 0$ and $S_{\text{ds}} \neq 0$). This matches the exam question verbatim.

INCORRECT **Option (d): Wrench fault**

Detailed Refutation: A *wrench fault* is a specific geological term (introduced by Kennedy, 1946, and Wilcox et al., 1973) denoting a high-angle to vertical regional strike-slip fault system where deformation is dominated by horizontal strike shear under an Andersonian stress regime where σ_2 is strictly vertical. While small localized vertical motions occur along wrench faults due to bends and stepovers (flower structures), wrench faults are kinematically strike-slip faults, not a generalized class defined by simultaneous dual components.



SYNTHESIS & HIGH-YIELD PRACTICE QUESTIONS

Curated Problems Covering Fault Trigonometry, Rake Calculations, Andersonian Regimes & Kinematic Analysis

High-Yield Exam Practice Questions

Practice Q1 (Numerical / Trigonometry): Net Slip Decomposition

A planar fault striking 090° and dipping 60° records slickenlines with a pitch (rake) of 30° measured from the eastern strike tip. Field measurement reveals a total net slip of 10 m . What are the magnitudes of the strike-slip component (S_{ss}) and the dip-slip component (S_{ds})?

Solution & Rationale:

Using standard vector projection:

- Strike-slip component: $S_{\text{ss}} = S \cdot \cos(\phi) = 10\text{ m} \cdot \cos(30^\circ) = 10 \cdot 0.8660 = 8.66\text{ m}$.
- Dip-slip component: $S_{\text{ds}} = S \cdot \sin(\phi) = 10\text{ m} \cdot \sin(30^\circ) = 10 \cdot 0.5000 = 5.00\text{ m}$.

Because both components are non-zero ($S_{\text{ss}} = 8.66\text{ m}$ and $S_{\text{ds}} = 5.00\text{ m}$), the structure is an oblique-slip fault.

Practice Q2 (Throw & Heave Calculation): Spatial Projections

For the fault in Practice Q1 ($S_{\text{ds}} = 5.00\text{ m}$), fault dip $\delta = 60^\circ$, calculate the vertical throw and horizontal heave produced by the displacement.

Solution & Rationale:

- $\text{Throw} = S_{\text{ds}} \cdot \sin(\delta) = 5.00\text{ m} \cdot \sin(60^\circ) = 5.00 \cdot 0.8660 = 4.33\text{ m}$ vertical displacement.
- $\text{Heave} = S_{\text{ds}} \cdot \cos(\delta) = 5.00\text{ m} \cdot \cos(60^\circ) = 5.00 \cdot 0.5000 = 2.50\text{ m}$ horizontal displacement perpendicular to strike.

Practice Q3 (Kinematics & Pitch): Pitch Limits for Oblique Slip

According to the standard classification scheme of Angelier (1994) adopted in modern structural geology textbooks (Fossen, 2010), faults are classified as strictly *oblique-slip* when the pitch (rake) ϕ of slickenlines falls within which range?

Answer: Between 10° and 80° (or in tripartite divisions, 15° to 75°). Faults with pitch $< 10^\circ$ are classified as strike-slip, while faults with pitch $> 80^\circ$ are classified as dip-slip.

Practice Q4 (Tectonic Regimes): Transpression Structures

In an oblique convergence zone where strike-slip shear is combined with regional shortening (transpression), what structural association is characteristically formed along the fault zone?

Answer: Positive Flower Structure (Palm-Tree Structure) consisting of upward-diverging reverse oblique-slip fault splays, creating structural pop-ups and en-echelon fold belts.

Practice Q5 (Tectonic Regimes): Transtension Structures

In an oblique divergence setting where strike-slip motion is accompanied by crustal extension (transtension), what basin geometry typically develops between releasing stepovers?

Answer: Pull-Apart Basin (or Rhomb Graben), bounded by normal oblique-slip faults exhibiting negative flower structures (tulip structures). Classic modern example: the Dead Sea basin along the Dead Sea Transform.

Practice Q6 (Anderson's Theory vs. Oblique Slip): Dynamic Limitations

Why does E. M. Anderson's (1905, 1951) standard dynamic classification of faults fail to predict oblique-slip faults?

Answer: Andersonian classification assumes that the Earth's surface is a shear-stress-free boundary ($\tau = 0$), forcing one of the three principal stress axes ($\sigma_1, \sigma_2, \sigma_3$) to be strictly vertical and the other two strictly horizontal. This theoretical constraint produces only pure normal, pure reverse, or pure strike-slip faults. Oblique-slip faults occur when faults reactivate pre-existing planes of weakness misoriented with respect to the principal stress axes, or in three-dimensional (triaxial / non-plane) strain fields.

Practice Q7 (Slickenlines & Asymmetries): Determining Kinematic Sense

What structural features on a fault plane allow a geologist to distinguish between the two possible senses of movement along an oblique slickenline?

Answer: Kinematic indicators (fault-surface asymmetries), including: (1) Asymmetric calcite/quartz mineral fiber steps (fibers grow in the shadow behind steps facing the movement direction of the opposite block), (2) Riedel shears (R, R', P shears), (3) Crescentic gouges and chatter marks, and (4) Stylolitic teeth oriented parallel to the maximum shortening direction.

Practice Q8 (Subtype Identification): Scenario Analysis

A mining engineer observes a fault dipping 50° to the west. Mining markers show that the hanging wall block moved downward relative to the footwall, and a displaced vertical dyke shows that the opposite block moved to the right of an observer facing the fault. What is the precise kinematic name of this fault?

Answer: Dextral-Normal Oblique-Slip Fault (or Right-lateral Normal Fault). The downward motion of the hanging wall constitutes the normal dip-slip component, while the rightward relative shift of the opposite block provides the dextral strike-slip component.

Comprehensive Kinematic Summary Matrix

Fault Classification	Primary Stress Axis Orientation (Andersonian)	Slickenline Pitch (ϕ)	Slip Vector Components	Displacement Effect on Planar Markers
Normal Fault	σ_1 vertical; σ_2, σ_3 horizontal	$\phi \approx 90^\circ$ (down-dip)	$S_{\text{ds}} > 0$ (hanging wall down), $S_{\text{ss}} = 0$	Vertical throw, horizontal crustal extension (heave), stratigraphy dropped.
Reverse / Thrust Fault	σ_3 vertical; σ_1, σ_2 horizontal	$\phi \approx 90^\circ$ (up-dip)	$S_{\text{ds}} > 0$ (hanging wall up), $S_{\text{ss}} = 0$	Vertical throw, crustal shortening, stratigraphy repeated.
Strike-Slip Fault	σ_2 vertical; σ_1, σ_3 horizontal	$\phi \approx 0^\circ$ (horizontal)	$S_{\text{ss}} > 0$ (dextral or sinistral), $S_{\text{ds}} = 0$	Lateral separation of vertical markers (dykes, rivers), negligible vertical throw.
Oblique-Slip Fault (The Answer)	Non-Andersonian (stress axes oblique or reactivated fabric)	$10^\circ \leq \phi \leq 80^\circ$ (oblique)	Both $S_{\text{ss}} \neq 0$ AND $S_{\text{ds}} \neq 0$	Combined vertical offset (throw) and lateral separation in a single slip event.